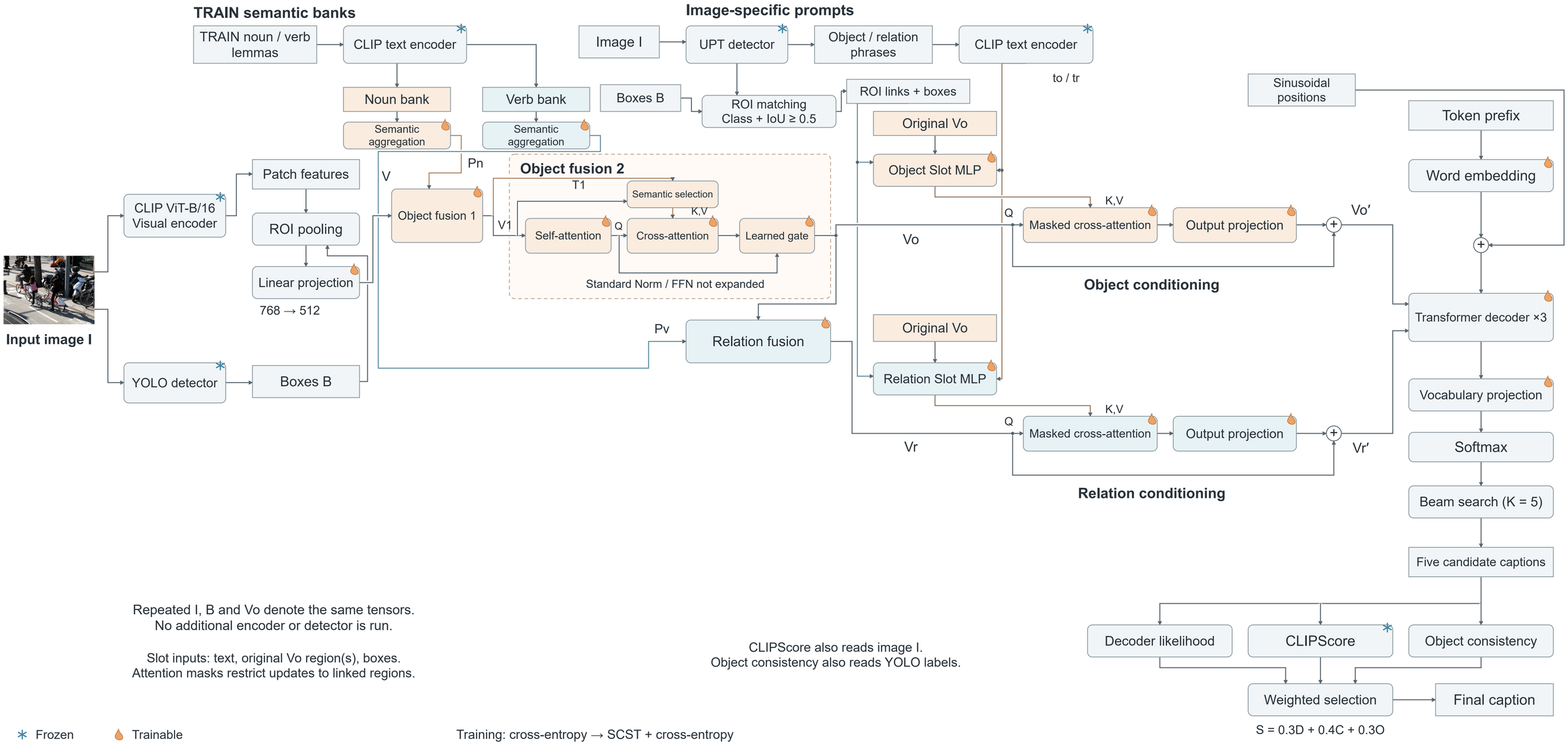

TRAIN semantic banks
TRAIN noun / verb lemmas
CLIP text encoder
Noun bank
Verb bank
Semantic aggregation
Semantic aggregation
Image-specific prompts
Image I
UPT detector
Object / relation phrases
CLIP text encoder
to / tr
Boxes B
ROI matching Class + IoU ≥ 0.5
ROI links + boxes
Original Vo
Object Slot MLP
Sinusoidal positions
Token prefix
Word embedding
Input image I
CLIP ViT-B/16 Visual encoder
Patch features
ROI pooling
Linear projection
768 → 512
V
Pn
Object fusion 1
Object fusion 2
T1
Semantic selection
K,V
V1
Self-attention
Q
Cross-attention
Learned gate
Standard Norm / FFN not expanded
Vo
Q
K,V
Masked cross-attention
Output projection
Vo′
Object conditioning
Transformer decoder ×3
Pv
Relation fusion
Original Vo
Relation Slot MLP
Vr
Q
K,V
Masked cross-attention
Output projection
Vr′
Relation conditioning
Vocabulary projection
Softmax
Beam search (K = 5)
YOLO detector
Boxes B
Five candidate captions
Decoder likelihood
CLIPScore
Object consistency
Weighted selection
Final caption
S = 0.3D + 0.4C + 0.3O
Repeated I, B and Vo denote the same tensors.
No additional encoder or detector is run.
Slot inputs: text, original Vo region(s), boxes.
Attention masks restrict updates to linked regions.
CLIPScore also reads image I.
Object consistency also reads YOLO labels.
Frozen
Trainable
Training: cross-entropy → SCST + cross-entropy